

Sripad Tiruvaipati

Data Analyst

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Summary

Results-driven Data Analyst with 3+ years of experience in Healthcare and Financial Services, leveraging SQL, Python, Power BI, Tableau, ETL, and Data Modeling to transform complex data into actionable business insights. Skilled in Business Intelligence, Data Visualization, KPI Reporting, Predictive Analytics, Generative AI, driving reporting automation, process optimization, and data-driven decision-making.

Technical skills

Languages:	Python, SQL
Packages/Libraries:	NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn, TensorFlow
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Database Management:	MySQL, PostgreSQL, SQL, dbt
Cloud Technologies:	AWS, Azure
Methodologies:	SDLC, Agile, Waterfall, Predictive Modeling
Version Control:	Git, GitHub
AI & Data Manipulation:	Predictive Analytics, Generative AI, Data Analysis, Data Mining, Data Preprocessing, Data Mapping, Data Cleaning, Data Visualization, Data Modeling, Data Warehousing, Data Storytelling, Data Wrangling, Data Integration, Data Transformation
Other Skills:	SSIS, SSRS, Machine Learning Algorithms, Prompt Engineering, CI/CD, Probability Distributions, Confidence Intervals, ANOVA, Advanced Analytics, ETL Processes, Informatica, Report Generation, Regression, A/B Testing, Forecasting & Modeling, Hypothesis Testing, Time Series Analysis, Root Cause Analysis

Work experience

Data Analyst Elevance Health, MA, USA Present	Sep 2024 –
- Led an AI-driven population health analytics initiative focused on patient risk stratification, care gap detection, readmission prevention, and value-based care optimization across multi-state healthcare populations. - Developed scalable data integration workflows using SQL and dbt to unify EHR, claims, pharmacy, laboratory, and enrollment data into longitudinal patient records, improving data reliability by 15% for 22K+ members. - Established enterprise data governance, validation, and quality assurance frameworks, implementing automated profiling and reconciliation controls that improved reporting accuracy to 98.4%. - Built predictive analytics, risk stratification, and propensity models using Python, Pandas, NumPy, and Scikit-learn to identify high-risk patients, predict avoidable hospitalizations, and reduce healthcare expenditures by 9%. - Designed 4 executive-facing Power BI dashboards using DAX, KPI scorecards, and row-level security, delivering real-time insights into risk scores, HEDIS measures, utilization trends, and rediscovers for 50+ stakeholders. - Utilized AWS S3 and Amazon Athena to perform large-scale healthcare analytics, enable secure HIPAA-compliant data access, and reduce data retrieval time by 20%. - Applied BigQuery partitioning, clustering, and query optimization techniques to analyze healthcare utilization patterns, identify cost drivers, and reduce turnaround time by 18%. - Collaborated with clinicians, care managers, executives, compliance leaders, and data engineers in Agile environments to deliver analytics solutions, automate KPI reporting, and improve quality and reimbursement outcomes.	

- Directed digital onboarding analytics for identity verification, KYC due diligence, sanctions screening, and AML risk assessments, ensuring compliance with **AML**, **KYC**, and **BSA/FinCEN** regulations while mitigating onboarding risk.

- Architected **Apache Spark** and **Azure Data Factory** pipelines processing **25GB+** onboarding, transaction, and identity verification records daily, enabling near real-time risk scoring, watchlist matching, sanctions screening, and fraud surveillance.
- Generated **90+** AML investigation packages and compliance reports using **Alteryx**, reducing manual effort by **22%** while improving audit readiness and regulatory reporting efficiency.
- Orchestrated **6+** executive **Tableau** dashboards tracking onboarding conversion, KYC completion, fraud alerts, customer risk segmentation, sanctions screening outcomes, and AML KPIs, reducing reporting latency by **18%**.
- Engineered fraud detection and customer risk-scoring models using **Scikit-learn**, **Pandas**, and **NumPy** to identify synthetic identities, mule accounts, account takeovers, and suspicious transactions, achieving **82%** anomaly detection accuracy.
- Optimized AML monitoring thresholds and onboarding risk controls through statistical validation and A/B testing using **SciPy**, improving suspicious activity detection precision by **7%** and reducing false-positive investigations.
- Streamlined KYC, AML, and fraud analysis using **Advanced Excel**, leveraging **Pivot Tables**, **VLOOKUP**, **XLOOKUP**, **Power Query**, and **INDEX-MATCH** to automate reconciliations, improve data accuracy, and accelerate compliance reporting by **20%**.
- Partnered with Fraud Operations, Compliance, Risk, Product, and Engineering teams in **Agile Scrum** environments to operationalize risk analytics, automate onboarding controls, and strengthen financial crime detection and regulatory reporting.

Projects

Credit Card Fraud Detection – Big Data Analysis

- **Analyzed 60,000+ credit card transactions** using **Python** and **SQL** to identify fraudulent activity patterns and anomalies.
- Performed **data cleaning**, **feature engineering**, and **exploration data analysis (EDA)** on **highly imbalanced transaction data**.
- Developed **machine learning models** to classify fraudulent transactions and evaluated performance using **Precision**, **Recall**, **F1-Score**, and **ROC-AUC**.
- Visualized fraud trends and risk indicators using **Python libraries** to support **data-driven decision-making**.

Brain Tumor Detection Using MRI Image Classification

- Developed a deep learning pipeline in **PyTorch** to classify brain MRI images into four tumor categories using a custom CNN architecture and transfer learning with **ResNet18**.
- Preprocessed and augmented MRI datasets through image normalization, resizing, and transformation techniques to improve model robustness and generalization.
- Designed and trained a multi-layer **CNN** with convolutional, pooling, dropout, and fully connected layers, achieving **99.5% classification accuracy** on test data.
- Evaluated model performance using **confusion matrices**, **precision**, **recall**, **F1-score**, and **classification reports** to validate prediction reliability.
- Compared custom **CNN** and pretrained **ResNet18** models, demonstrating the effectiveness of transfer learning for medical image classification tasks.

Education

Master of Science in Data Science- University of Massachusetts Dartmouth, MA, USA.

Bachelor of Technology in Electronics and Communication Engineering - Vasavi College of Engineering, Hyderabad, India.

Certification

- **AWS certified cloud practitioner**